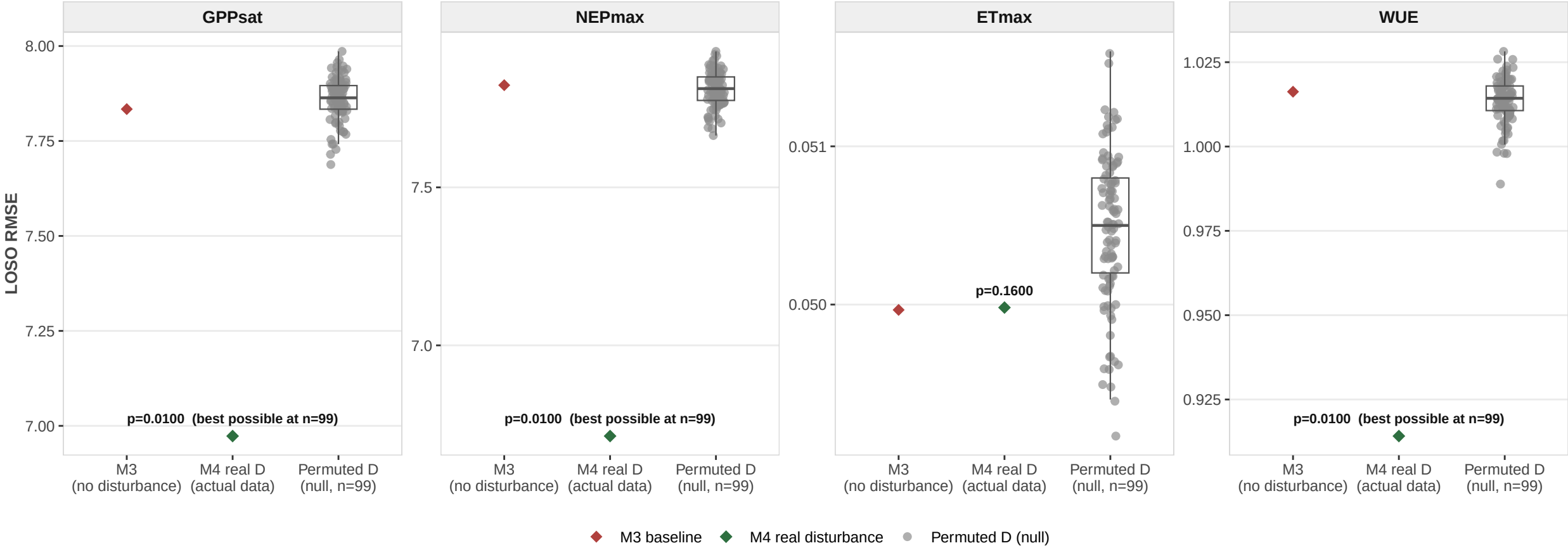
Permutation test: is the disturbance−block RMSE gain real information, or added flexibility?

XGBoost−Optuna, tc>=30 (93 sites / 395 site−years), 12−month window, LOSO CV | Null = M4 retrained with the disturbance block's rows block−permuted (99 reps)



p = (# permuted reps with RMSE <= real M4 RMSE + 1) / (n_perm + 1), one−sided. Floor at n_perm=99 is p=0.0100 (real M4 beating all 99/99 permutations).
ETmax's real M4 falling inside the null cloud is expected: disturbance did not improve ETmax in the main results either − an internal negative control.